

RIFT: Future-Recovery Routing for Privileged On-Policy Self-Distillation

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Abstract

On-policy self-distillation (OPSD) trains reasoning models on their own trajectories using a solution-conditioned view of the same model as dense token-level supervision. Privileged correction is not uniformly beneficial: at a reasoning fork, a conflict may resolve along the student’s continuation or persist and require intervention. Existing current-state routing signals do not directly distinguish these cases. We introduce Recovery-Informed Forked Training (RIFT), a token-level framework that measures future reconvergence on the realized student rollout. Following the fixed-initial-policy implementation used in OPSD experiments, RIFT evaluates the same frozen initial policy with and without a verified solution. At high-entropy sign conflicts, it routes recoverable cases to the unprivileged reference distribution and persistent cases to the privileged distribution. A per-trajectory exact-rank rule enforces the prescribed local routing budget even under tied recovery scores. We further introduce REGAP, a fixed-prefix, fixed-action counterfactual protocol that separates privileged-context advantage, branch-action value, and transferable recovery skill. Under a unified 50-step Avg@12 evaluation on AIME24, AIME25, and HMMT25, RIFT averages 63.06 across two clean Qwen3-4B training seeds, compared with 61.67 for budget-matched OPSD, a 1.39 percentage-point improvement with a hierarchical 95% confidence interval of [0.23, 2.55]. A registered seed-44 comparison additionally yields a 2.41-point gain. Combining future recovery with current-state covariates improves AUPRC by 0.01191 over current covariates alone (95% CI [0.0049, 0.0188], $p = 0.0018$). Exact-rank routing achieves zero per-trajectory budget error and zero tie excess. These results show that future recovery provides an effective temporal signal for privileged-context arbitration while making token-level routing exact and reproducible.

Introduction

Knowledge distillation transfers a teacher’s predictive distribution to a student (Hinton, Vinyals, and Dean 2015; Kim and Rush 2016); same-capacity and iterative variants further show that distillation can serve as a learning procedure rather than only compression (Furlanello et al. 2018; Xie et al. 2020). For autoregressive models, training only

on fixed teacher trajectories creates a state-distribution mismatch, the classical motivation for on-policy imitation (Ross, Gordon, and Bagnell 2011). Generalized KD and MiniLLM address this mismatch by distilling on student-generated sequences (Agarwal et al. 2024; Gu et al. 2024). On-policy self-distillation (OPSD) makes the approach especially attractive for reasoning by evaluating the same model under student and privileged solution-conditioned contexts. Its experiments stabilize the target by fixing the privileged view to the initial policy, which supplies dense token-level supervision along the student’s realized trajectory (Zhao et al. 2026). The quality of this supervision depends on when privileged information should control a local reasoning decision.

Reasoning trajectories are not locally monotone. At a high-entropy fork, a student token may conflict with an answer-conditioned teacher and still lead to verification, backtracking, or a later return to the correct direction. Self-taught reasoning, verbal reflection, tree search, and adaptive test-time compute all exploit the fact that later computation can revise an earlier decision (Zelikman et al. 2022; Shinn et al. 2023; Yao et al. 2023; Snell et al. 2024). Overriding every such conflict treats privileged hindsight as immediate action credit and can suppress productive exploration (Kaur et al. 2026; Peng et al. 2026). Existing approaches reweight positions, decompose privileged targets, preserve hidden transitions, or blend teacher and student distributions (Xie et al. 2026; Shen et al. 2026; Li et al. 2026b,a). These methods primarily characterize the current token. The missing decision variable is temporal: does the student’s continuation recover from the present conflict, or does the disagreement persist?

We cast this decision as *token-level privilege arbitration* and introduce Recovery-Informed Forked Training (RIFT). During training, RIFT evaluates future reconvergence on the student’s already realized rollout. It first identifies high-entropy sign conflicts where the solution-conditioned view of the frozen initial policy suppresses the sampled student token but its unprivileged view does not. It then measures whether student-privileged-view Jensen–Shannon divergence decreases within a fixed future window. Recoverable conflicts are routed to the unprivileged reference distribution q^0 , preserving dense supervision without answer-conditioned override; persistent conflicts retain the privileged distribution q^+ . The routing signal is used only during training, so the resulting student incurs no reference-model

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or look-ahead cost at inference time.

Recovery routing also requires precise budget control. Thresholded quantiles can select an unpredictable number of tokens when recovery scores tie, coupling the proposed signal to a changing amount of intervention. RIFT resolves this problem with deterministic per-trajectory exact ranking. Recovery score, token position, and a fixed hash define a total order, and the router selects exactly the prescribed number of candidates on every trajectory. This construction makes route allocation reproducible and separates signal quality from training budget.

The same temporal evidence raises a second question: what does recovery identify? A privileged teacher may rescue a fixed branch even when that branch is a poor action for the student, and successful privileged continuation does not establish that the underlying skill transfers to a student without the privileged context. We introduce Counterfactual Rescue Gap analysis (RE_{GAP}) to separate three quantities: privileged-context advantage, branch-action advantage, and transferable recovery skill. RE_{GAP} fixes the prefix and branch action, adds matched neutral controls, and varies access to privileged information. This analysis motivates RIFT’s central design choice: recovery controls the *source of supervision*; it is not used as direct reward for the sampled action.

Our experiments provide three complementary forms of evidence. First, future recovery alone reaches AUPRC/AUROC of 0.9098/0.6276, compared with 0.9065/0.6251 for current covariates, and combining both reaches 0.9184/0.6320. The combined predictor significantly improves both AUPRC and AUROC over current covariates alone. Second, exact-rank routing achieves zero per-trajectory budget error and zero tie excess. Third, under a unified 50-step Avg@12 evaluation, RIFT improves over matched OPSD in both clean Qwen3-4B replications: 689 versus 678 correct completions at seed 42 and 673 versus 654 at seed 43. The clean-seed mean gain is 1.39 points, with a hierarchical 95% interval of [0.23, 2.55]. Together, these results connect a temporally defined routing signal to exact intervention control and improved budget-matched reasoning performance.

Our contributions are:

- We formulate privileged supervision as token-level arbitration and introduce future recovery on the realized student rollout as its temporal decision signal.
- We develop RIFT, a context-conditioned target-routing framework that assigns recoverable and persistent conflicts to the unprivileged and privileged views of the same frozen initial policy, respectively.
- We introduce deterministic exact-rank routing, which enforces the prescribed per-trajectory route count under tied scores and makes the intervention reproducible.
- We develop RE_{GAP}, a fixed-prefix, fixed-action counterfactual protocol that distinguishes privileged-context advantage, action advantage, and transferable recovery skill.
- We demonstrate consistent 50-step improvements over matched OPSD in two clean Qwen3-4B training-seed replications, with a 1.39-point pooled gain, together

with predictive, calibration, target-source, cross-scale, and counterfactual analyses of the mechanism.

Related Work

Distillation on Student-Induced Distributions

Knowledge distillation was introduced as distribution matching between a teacher and a deployable student (Hinton, Vinyals, and Dean 2015) and was later extended to sequence-level targets, policies, and intermediate representations (Kim and Rush 2016; Rusu et al. 2015; Sun et al. 2019). Same-capacity students and iterative self-training further established distillation as a mechanism for improving the learner itself, beyond model compression (Furlanello et al. 2018; Xie et al. 2020). These methods typically define supervision on a fixed data distribution.

Sequential prediction changes the states on which imitation must succeed: an early student action determines the prefixes encountered later. DAgger addresses this covariate shift by collecting supervision under learner-induced states (Ross, Gordon, and Bagnell 2011). GKD and MiniLLM bring the same principle to autoregressive language models by querying the teacher on student-generated sequences (Agarwal et al. 2024; Gu et al. 2024). OPSD further uses a frozen self-teacher with privileged reasoning context, such as a verified solution, to supervise the student’s own rollout in its experimental fixed-initial-policy implementation (Zhao et al. 2026). RIFT operates within this on-policy setting and addresses a finer decision: which contextual view of that initial policy should supervise an individual reasoning fork.

Controlling Privileged Reasoning Supervision

Process supervision assigns feedback to intermediate reasoning steps rather than only the final answer (Uesato et al. 2022; Lightman et al. 2023). OPSD provides an alternative dense signal by conditioning a teacher distribution on a verified solution, but answer-conditioned supervision can alter local credit at uncertain forks (Kaur et al. 2026; Peng et al. 2026). Recent methods control this influence through three broad intervention families.

The first family changes *where* privileged supervision is emphasized. Position-aware OPD discounts tokens according to accumulated discrepancy, while ReNIO reweights complete trajectories using token-probability ratios (Xie et al. 2026; Lin, Chen, and Zhang 2026). The second changes *what target* is distilled. PHF supervises privileged hidden transitions; RLCSD contrasts correct and incorrect privileged hints; Purified OPSD and AR-OPD separate a locally reachable component from a privileged residual (Li et al. 2026b; Pan et al. 2026; Shen et al. 2026; Zhang 2026). DemoPSD blends teacher distributions using current disagreement, whereas AD-OPSD anchors actively suppressed high-entropy tokens to a frozen reference prior (Li et al. 2026a; Peng et al. 2026). The third uses continuation information: Hindsight Self-Distillation localizes credit using a successful peer continuation (Li, Hong, and Lan 2026).

RIFT combines a distinct temporal signal with context-conditioned target switching. It measures future reconvergence on the same realized student rollout, ranks candidate

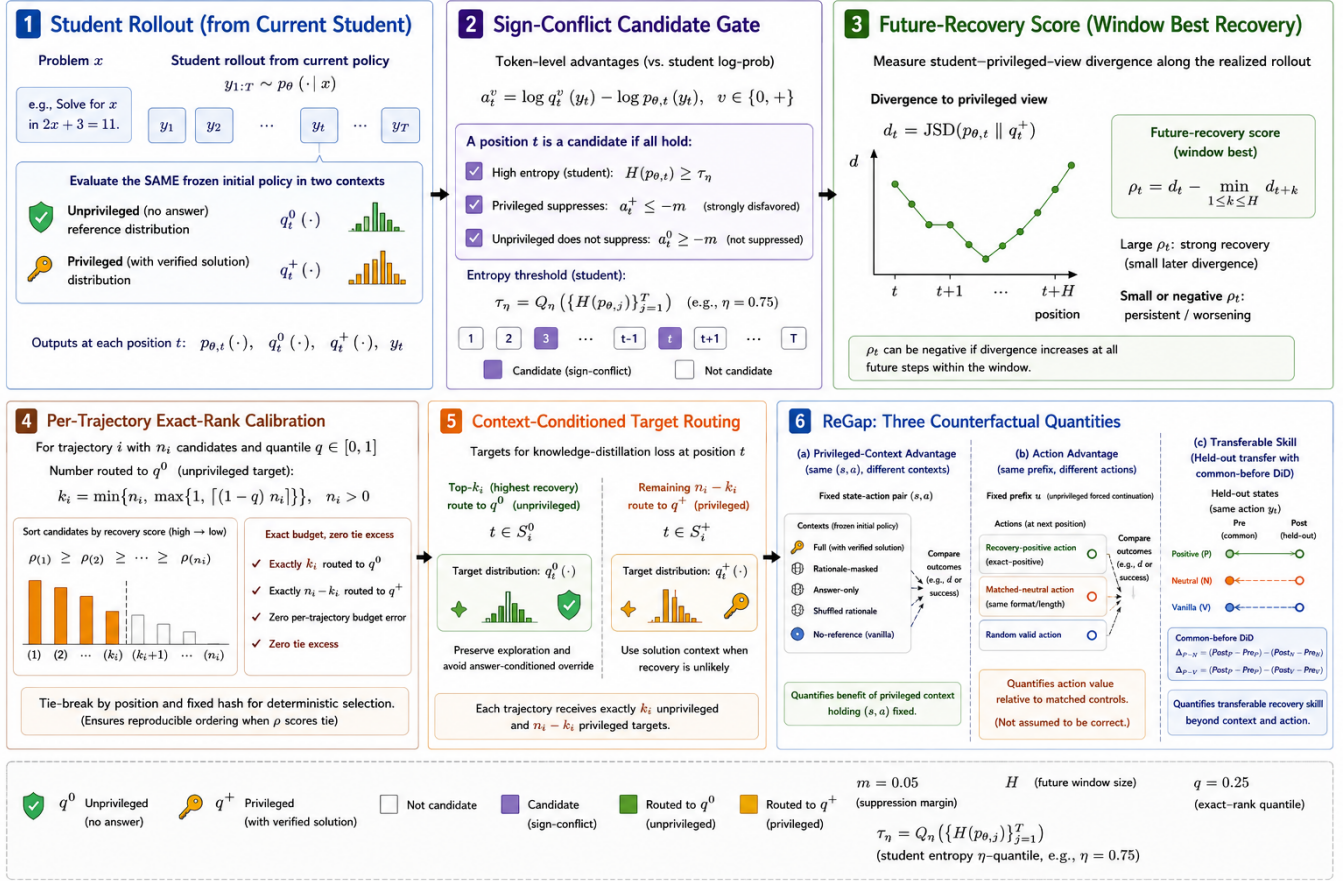


Figure 1: Overview of RIFT and REGAP. RIFT generates an on-policy student rollout, filters sign-conflict candidates, scores window-best future recovery, and applies deterministic per-trajectory exact ranking before routing recoverable and persistent conflicts to the unprivileged q^0 and privileged q^+ views of the same frozen policy. REGAP separately tests privileged-context advantage, branch-action advantage, and held-out transferable recovery. Routing and future information are training-only.

sign conflicts at an exact local budget, and switches recoverable tokens from the privileged distribution q^+ to the unprivileged reference distribution q^0 of the same frozen initial policy. Persistent conflicts retain q^+ . This design preserves dense supervision while separating future evidence from the amount of routed training signal.

Recovery, Search, and Transfer

Several reasoning paradigms exploit information that becomes available after an initial decision. STaR learns from self-generated rationales, Reflexion uses later feedback as verbal memory, Tree of Thoughts explicitly explores and backtracks, and adaptive test-time compute allocates additional search to difficult problems (Zelikman et al. 2022; Shinn et al. 2023; Yao et al. 2023; Snell et al. 2024). These results motivate recovery as a property of a trajectory rather than a single token. RIFT turns that property into a training-time quantity: the future decrease in student-privileged-reference divergence determines the local source of supervision, while deployment uses the trained student alone.

Recovery also creates an identification problem. A privileged teacher may complete a fixed branch because it observes the answer, even when the branch itself is not a better student action; successful privileged continuation also does not establish transfer to the no-reference student. Context-return experiments expose this distinction between privileged performance and robust internalization (Wang et al. 2026). REGAP formalizes it through matched fixed-prefix, fixed-action comparisons that separately estimate privileged-context advantage, branch-action advantage, and transferable recovery skill.

Recovery-Informed Forked Training

Problem Setup

Let x denote a problem, r a verified solution, and $y = (y_1, \dots, y_T)$ an on-policy student rollout. At position t , the trainable student distribution is

$$p_{\theta,t} = p_{\theta}(\cdot | x, y_{<t}). \quad (1)$$

OPSD is defined by applying one model under student and privileged contexts. Its experimental fixed-teacher implementation holds the target at the initial policy while the

student LoRA parameters update. We adopt that implementation: $\bar{\theta} = \theta_0$ denotes the frozen initial policy and θ denotes the current student parameters. We evaluate $p_{\bar{\theta}}$ under two contexts. The privileged reference distribution observes the verified solution,

$$q_t^+ = q_{\bar{\theta}}(\cdot \mid x, r, y_{<t}), \quad (2)$$

whereas the unprivileged reference distribution receives the inference-time context only,

$$q_t^0 = q_{\bar{\theta}}(\cdot \mid x, y_{<t}). \quad (3)$$

Thus q^+ and q^0 share one frozen parameter set and differ only in conditioning context. They are not separate teacher models. In a dynamic shared-current-weight formulation, $q_t^0 = p_{\theta,t}$ and routing to q^0 would yield zero divergence; RIFT instead follows the fixed-initial-policy implementation used by the OPSD experiments. Standard OPSD distills the privileged distribution densely along the rollout:

$$\mathcal{L}_{\text{OPSD}} = \frac{1}{T} \sum_{t=1}^T \text{JSD}(p_{\theta,t} \parallel q_t^+). \quad (4)$$

The rollout is on-policy and the target is conditioned on hindsight. RIFT retains the same rollout and frozen initial policy while making its conditioning context—privileged or unprivileged—a token-level routing decision.

Sign-Conflict Candidates

For the sampled token y_t , define the privileged and unprivileged log-support margins relative to the student:

$$a_t^+ = \log q_t^+(y_t) - \log p_{\theta,t}(y_t), \quad (5)$$

$$a_t^0 = \log q_t^0(y_t) - \log p_{\theta,t}(y_t). \quad (6)$$

A token is a routing candidate when the student is uncertain, the privileged view suppresses the sampled action, and the unprivileged view does not reject it by the same margin:

$$C_t = \mathbb{K}[H(p_{\theta,t}) \geq Q_{\eta}(H) \wedge a_t^+ \leq -m \wedge a_t^0 \geq -m]. \quad (7)$$

Here $Q_{\eta}(H)$ is the within-trajectory entropy quantile and m is the sign margin. The formal recipe uses $\eta = 0.75$ and $m = 0.05$.

Future Recovery on the Realized Rollout

Let

$$d_t = \text{JSD}(p_{\theta,t} \parallel q_t^+) \quad (8)$$

measure current divergence from the privileged reference. For a look-ahead window H , the recovery score is

$$\rho_t = d_t - \min_{1 \leq k \leq H} d_{t+k}. \quad (9)$$

A large ρ_t means that a large disagreement at t falls later on the same student continuation. A small score indicates persistent divergence. We use $H = 32$ valid future tokens in the primary configuration. Positions without a full valid future window are excluded from the primary candidate set.

Equation 9 is an observed-rollout hindsight statistic. It does not estimate the counterfactual value of every alternative action. This distinction motivates the separate **ReGAP** analysis below.

Deterministic Exact-Rank Routing

For trajectory i , let $\mathcal{C}_i$ be its candidate set and $n_i = |\mathcal{C}_i|$. Given quantile parameter q , the prescribed route count is

$$k_i = \begin{cases} \min\{n_i, \max\{1, \lceil (1-q)n_i \rceil\}\}, & n_i > 0, \\ 0, & n_i = 0. \end{cases} \quad (10)$$

Candidates are sorted lexicographically by decreasing ρ_t , increasing token position, and a fixed integer hash. The first k_i receive $g_t = 1$; all others receive $g_t = 0$. The primary $q = 0.25$ configuration therefore selects approximately the highest-recovery 75% of candidates while maintaining an exact deterministic count.

A threshold quantile instead applies

$$g_t = \mathbb{K}[t \in \mathcal{C}_i \wedge \rho_t \geq Q_q(\{\rho_j : j \in \mathcal{C}_i\})]. \quad (11)$$

Quantized ties make this count data-dependent. Exact rank separates the choice of *how many* tokens to route from the values and ties of their scores.

Context-Conditioned Target Switching

The routed loss is

$$\mathcal{L}_{\text{RIFT}} = \frac{1}{T} \sum_{t=1}^T \left[(1 - g_t) \text{JSD}(p_{\theta,t} \parallel q_t^+) + g_t \text{JSD}(p_{\theta,t} \parallel q_t^0) \right]. \quad (12)$$

Thus, routing does not mean removing the token loss. A routed conflict receives dense supervision from the unprivileged view of the frozen initial policy; an unrouted token retains the privileged view. The unprivileged distribution acts as a stable reference rather than a trainable self-target.

Training and inference cost. The student rollout and both contextual reference distributions are needed during training. Future recovery is computed after the rollout has been generated, so it does not add inference-time look-ahead. At deployment, RIFT is an ordinary fine-tuned student with no teacher, route classifier, verifier, or future window.

Implementation invariants. Every formal run records candidate count, target route count, actual route count, boundary ties, per-trajectory budget error, candidate checksum, and route-mask checksum. A formal run is accepted only if all steps are finite and the exact-budget error is zero.

ReGAP: Counterfactual Rescue Is Not Action Credit

Let $s = (x, y_{<t})$ be a fixed student state, $a = y_t$ a fixed branch action, and c a problem-aligned privileged rationale. Let $V_T(s, a, c)$ and $V_S(s, a)$ denote verifier success probabilities under privileged-teacher and unprivileged-student continuations. The exact rescue gap is

$$G_{\text{exact}}(s, a) = V_T(s, a, c) - V_S(s, a). \quad (13)$$

The gap asks whether privileged context can rescue the same state-action pair. It does not, by itself, answer whether a was the right branch or whether the student can internalize the rescue.

Three Identified Quantities

Privileged-context advantage. We hold (s, a) fixed and compare full problem-aligned context with rationale-masked, answer-only, shuffled, and no-reference context conditions. This identifies whether rescue depends on aligned privileged information rather than a changed action or an answer leak.

Action advantage. We compare an exact-positive branch with matched-neutral and random valid branches. Matching uses problem, prefix position, original NLL/JSD, length, and difficulty. Forced continuations from the same prefix isolate whether a rescuable action is better for the unprivileged student.

Transferable skill. We train on exact-positive recovery, matched-neutral recovery, and vanilla arms, then evaluate unprivileged behavior on problem-disjoint held-out states. The primary analysis uses direct positive-neutral and positive-vanilla contrasts plus difference-in-differences relative to a common pre-training measurement.

The logic is

$$\begin{aligned} G_{\text{exact}} > 0 &\nRightarrow \text{better student action,} \\ \text{better student action} &\nRightarrow \text{transferable rescue.} \end{aligned} \quad (14)$$

REGAP therefore supplies a boundary analysis for RIFT’s target-source routing: privileged solvability is evidence to analyze, not an automatic local reward.

Experimental Setup

Training Protocol

The primary experiments use Qwen3-4B (Yang et al. 2025) and a frozen 29,434-example OpenThoughts Math 30K OPSD corpus (Zhao et al. 2026; Guha et al. 2025); cross-scale tests use Qwen3-1.7B/8B without router retuning. The student is adapted with LoRA rank 64 and scale 128 over the attention and feed-forward projections. Both reference distributions use the same frozen initial policy and differ only in access to the verified solution. All same-budget primary methods use 50 optimizer steps, learning rate 5×10^{-6} , 1,024-token on-policy training rollouts, temperature 1.1, top- p 0.95, top- k 20, and the same data order within a registered training seed. The primary RIFT configuration uses entropy quantile 0.75, sign margin 0.05, recovery window 32, and exact-rank $q = 0.25$.

Evaluation

We evaluate frozen 30-problem snapshots of AIME24, AIME25, and HMMT25 (Hugging Face H4 2025; yentinglin 2025; MathArena 2025), with 12 completions per problem. A frozen `math-verify` v0.8.0 evaluator (Kydliček 2024) scores the final balanced `\boxed{\dots}` expression. Each method therefore produces 360 records per benchmark and 1,080 across the triad; Avg@12 is the fraction correct. Decoding settings and exact asset digests are in the supplement. All counts use the paper snapshot frozen on 2026-07-23.

Baselines and Controls

The 50-step comparison includes matched OPSD, three exact-budget random masks, Current-JSD routing, a zero-loss target ablation, AD-OPSD (Peng et al. 2026), and DemoPSD (Li et al. 2026a). Signal analysis compares entropy, position, current JSD, sampled-token suppression margin, AD risk, past recovery, shuffled future, true future recovery, and current covariates with or without future recovery.

Signal and Counterfactual Data

The forced-continuation study contains 512 candidate branches from 128 problems, with four fixed continuation seeds per branch (2,048 continuations). The robust-recovery label has 442 positives among 512 candidates (86.33%). Predictors are evaluated out of fold. Precision@25% uses equal selection coverage, and uncertainty is clustered by problem.

The REGAP transfer study uses three training seeds. Each seed contains 24 complete records from the same 17 problem clusters, evaluated under exact-positive, matched-neutral, and vanilla exposures. Balance checks attain a maximum absolute standardized mean difference of 0.02385, and each replay arm exposes 10% of eligible tokens.

Statistical Reporting and Provenance

Problems, rather than individual stochastic completions, are the unit of inference. We preserve completion-level raw records, evaluator versions, training and evaluation seeds, model/data hashes, commands, manifests, and artifact checksums. All numerical results in this paper follow a single result snapshot frozen on 2026-07-23.

Results

Budget-Matched Reasoning Performance

Figure 2 reports the controlled 50-step comparisons. At training seed 42, RIFT obtains 689/1,080 (63.80), compared with 678/1,080 (62.78) for matched OPSD, a gain of 1.02 points. At seed 43, RIFT obtains 673/1,080 (62.31), compared with 654/1,080 (60.56), a gain of 1.76 points. Averaged over the two clean seeds, RIFT scores 63.06 versus 61.67 for matched OPSD; the 1.39-point gain has a hierarchical 95% interval of [0.23, 2.55]. The registered seed-44 comparison provides a third positive replication, with 676/1,080 (62.59) for RIFT and 650/1,080 (60.19) for matched OPSD. Among the evaluated 50-step methods, RIFT has the highest displayed Avg@12 point estimate.

Current-JSD routing scores 60.19, below the three-mask random-routing aggregate of 61.76 by 1.57 points. This control indicates that immediate disagreement magnitude alone is not a sufficient routing rule. Its supplied paired interval [-3.20, 0.09] includes zero ($p = 0.0718$).

What Does Future Recovery Predict?

Figure 3(a) summarizes the corrected signal-validity results. Future recovery alone reaches AUPRC 0.90980, a positive but non-significant increment of 0.00331 over current covariates alone (95% CI [-0.0012, 0.0078], $p = 0.149$). Combining future recovery with current covariates reaches AUPRC

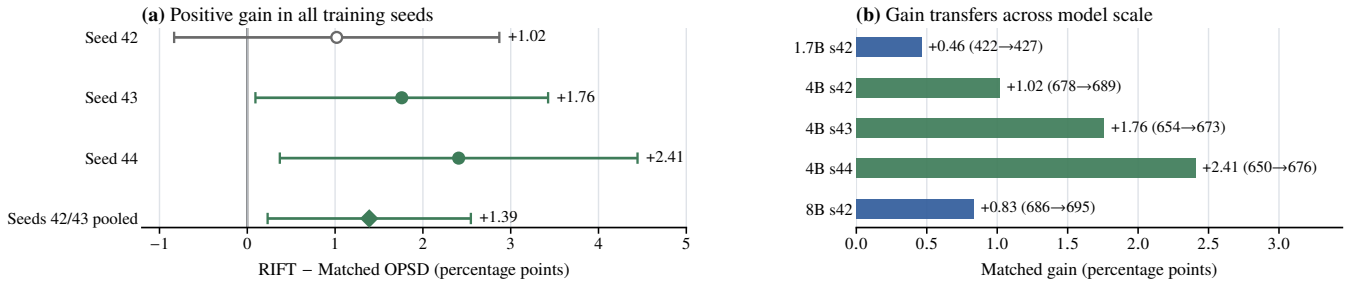


Figure 2: Matched 50-step performance. (a) Points show seed-specific gains with problem-cluster 95% intervals; the diamond hierarchically pools clean seeds 42/43. (b) Positive gains transfer from Qwen3-1.7B through Qwen3-8B without retuning the router; parenthetical labels give matched-to-RIFT correct-count transitions out of 1,080. Exact benchmark counts and all controls are in the supplement.

0.91840 and improves over current covariates by 0.01191 (95% CI [0.0049, 0.0188], $p = 0.0018$). The AUROC analysis shows the same pattern: future recovery alone changes AUROC by 0.00249 ($p = 0.42$), whereas the combined predictor improves AUROC by 0.00689 (95% CI [0.0015, 0.0123], $p = 0.013$). Thus future recovery contributes statistically detectable information when combined with current-state covariates, while its standalone increment is not significant.

The positive prevalence is 86.33%, so the prevalence-matched random AUPRC baseline is already approximately 0.8633. The corrected future-recovery AUPRC of 0.90980 exceeds this demanding baseline, while its AUROC of 0.62760 confirms positive ranking quality. We report both metrics to separate prevalence effects from discrimination.

Exact Rank Controls the Local Budget

Exact-rank routing is the only evaluated calibration rule with zero tie excess and exact local count by construction. It also has the lowest displayed route-fraction coefficient of variation, 0.04276, compared with 0.04791 for a per-trajectory threshold and 0.05308 for a fixed threshold. This result matters independently of downstream accuracy: it makes the intervention count reproducible under quantized scores and isolates routing quality from accidental budget changes. The complete calibration table is in the supplement; Figure 3(b) visualizes the joint calibration tradeoff.

Context-Conditioned Target Arbitration

With the token partition fixed, recoverable $\rightarrow q^0$ and persistent $\rightarrow q^+$ scores 691/1,080, versus 681/686/679 for uniform q^0 , uniform q^+ , and the reversed assignment (Figure 3c). The full descriptive comparison is in the supplement.

ReGap Separates Recovery Transfer from Matched Controls

Across three seeds, positive exposure lowers Branch NLL versus matched neutral by 1.361×10^{-4} (95% CI [1.439×10^{-5} , 2.632×10^{-4}], $p = 0.0285$) and recovery JSD by 2.242×10^{-5} (95% CI [1.323×10^{-5} , 3.221×10^{-5}], $p < 0.0001$). Recovery JSD also improves versus vanilla; Branch NLL has the intended direction but does not exclude

zero. Thus three of four registered pooled contrasts pass (Figure 4a); the full numerical table is in the supplement.

As an exploratory boundary analysis, Figure 4(b) evaluates one completion on each of 90 problems at longer generation budgets. RIFT scores 6.67/28.89/53.33 Pass@1 at 4k/8k/16k, compared with 4.44/22.22/47.78 for the base model. We treat this small diagnostic as descriptive rather than as a second primary endpoint.

Discussion

Future recovery is most useful jointly with current covariates, supporting a temporal complement to current-state routing rather than a standalone oracle. Exact rank adds a deterministic local invariant: score ties cannot change the number of routed candidates, although candidate prevalence may still vary. Finally, ReGap prevents privileged rescue from being read as local action credit. Positive exposure separates from matched neutral on both endpoints and from vanilla on recovery JSD; Branch NLL versus vanilla remains a consistent but non-significant trend. Our primary empirical claim is therefore restricted to the controlled 50-step Qwen3-4B setting, where all three matched training seeds favor RIFT.

Limitations

Future recovery is an observed-rollout statistic, not the counterfactual value of the sampled token; it depends on window length, stochasticity, and sequence censoring. The registered $q = 0.25$, $H = 32$, and candidate thresholds are not universal constants. The unprivileged target is a frozen reference rather than unconstrained exploration. Target-source arbitration is descriptive because it lacks paired uncertainty, and one of four ReGap pooled contrasts does not exclude zero.

Conclusion

Privileged supervision should depend on the temporal structure of teacher-student disagreement. RIFT operationalizes this principle by using future reconvergence on the student’s own rollout to switch between the privileged and unprivileged views of the same frozen initial policy. Exact-rank selection makes this intervention locally exact and reproducible. Under a unified 50-step Qwen3-4B protocol, RIFT improves over matched OPSD in both clean replications, with

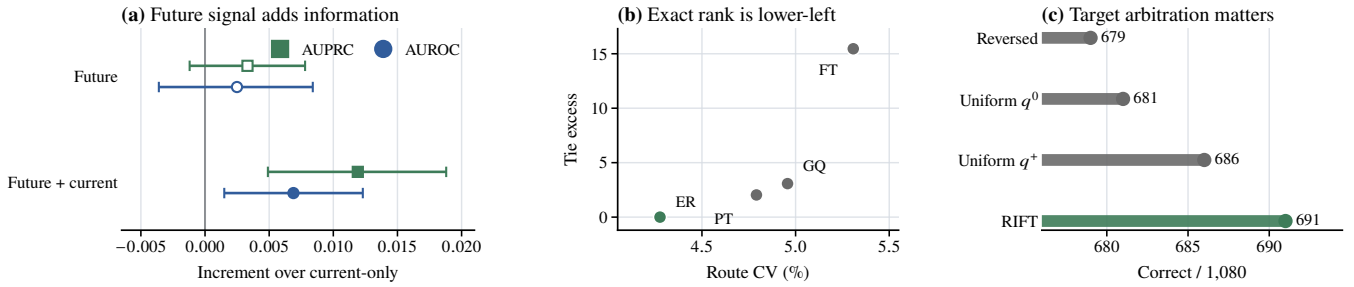


Figure 3: Routing diagnostics. (a) Problem-cluster 95% intervals for the increment over current-only prediction; filled markers exclude zero. (b) Calibration rules plotted by route-fraction CV and tie excess, where lower-left is better. Exact rank attains the lowest CV and zero tie excess. FT, GQ, PT, and ER denote fixed threshold, global q_{25} , per-trajectory threshold, and exact-rank q_{25} . (c) Correct counts for context-conditioned target policies under the same token partition and training budget.

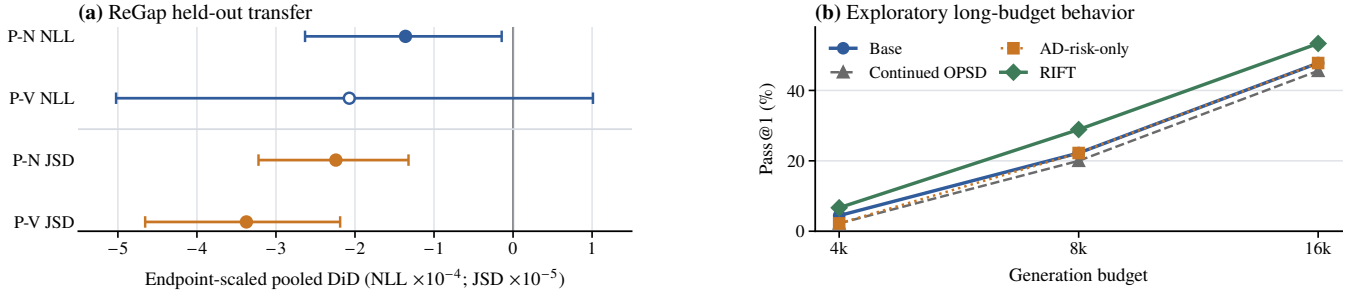


Figure 4: Transfer and horizon diagnostics. (a) ReGAP pooled difference-in-differences; NLL and JSD use the displayed endpoint-specific scales, lower is better, and filled markers exclude zero. P-N and P-V denote positive-minus-neutral and positive-minus-vanilla. (b) Exploratory Pass@1 over 90 problems at increasing generation budgets; these single-sample estimates are not used for the primary claim.

a mean gain of 1.39 points and a hierarchical 95% interval of [0.23, 2.55]. It also achieves the best displayed 50-step point estimate among the evaluated methods. ReGAP complements the training method by separating privileged rescue from action credit and transferable competence; three of four registered pooled contrasts pass across three training seeds. Together, the two components provide a training rule and an identification framework for deciding when a student should trust a privileged context-conditioned target.

Reproducibility Statement

Formal runs record the base model, data digest, frozen code snapshot, Docker image, command, environment, training and evaluation seeds, checkpoint hash, completion-level metrics, and a recursive artifact checksum. Exact-rank runs add per-step candidate and route counts, boundary ties, local budget errors, and candidate/route checksums. The main evaluation contains 1,080 records per method; the anonymous supplement provides the extended tables and protocol details used by the reproducibility checklist.

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